GATE ECE

Previous Year Questions — Chapter-wise, with Answers and Solutions

Typeset edition

Papers included: GATE 2020.

GATE 2020

GATE 2020 | GA Q.1 | 1 mark

The untimely loss of life is a cause of serious global concern as thousands of people get killed_____ accidents every year while many other die_____ diseases like cardio vascular disease, cancer, etc.

(A) in, of
(B) from, of
(C) during, from
(D) from, from

Answer: (A) [answer matches official key]

Solution. Step 1. People get killed *in* accidents and die *of* diseases: the idiomatic pair is *in, of*.

GATE 2020 | GA Q.2 | 1 mark

He was not only accused of theft_____ of conspiracy.

(A) rather
(B) but also
(C) but even
(D) rather than

Answer: (B) [answer matches official key]

Solution. Step 1. The correlative construction is *not only . . . but also*.

GATE 2020 | GA Q.3 | 1 mark

Select the word that fits the analogy: *Explicit : Implicit :: Express :* _____

(A) Impress
(B) Repress
(C) Compress
(D) Suppress

Answer: (B) [answer matches official key]

Solution. Step 1. *Implicit* is the opposite of *explicit*; the opposite of *express* (to show) is *repress* (to hold back).

GATE 2020 | GA Q.4 | 1 mark

The Canadian constitution requires that equal importance be given to English and French. Last year, Air Canada lost a lawsuit, and had to pay a six-figure fine to a French-speaking couple after they filed complaints about formal in-flight announcements in English lasting 15 seconds, as opposed to informal 5 second messages in French. The French-speaking couple were upset at _____.

(A) the in-flight announcements being made in English.
(B) the English announcements being clearer than the French ones.
(C) the English announcements being longer than the French ones.

(D) equal importance being given to English and French.

Answer: (C) [answer matches official key]

Solution. Step 1. The complaint compares a 15 s English announcement with a 5 s French one.
Step 2. They were upset that the English announcements were longer.

GATE 2020 | GA Q.5 | 1 mark

A superadditive function $f(\cdot)$ satisfies the following property $f(x_1 + x_2) \geq f(x_1) + f(x_2)$. Which of the following functions is a superadditive function for $x > 1$?

(A) e^x
(B) $\sqrt{x}$
(C) $1/x$
(D) e^{-x}

Answer: (A) [answer matches official key]

Solution. Step 1. Test e^x: $e^{x_1+x_2} = e^{x_1}e^{x_2} \geq e^{x_1} + e^{x_2}$ holds for $x_1, x_2 > 1$ (since $(a-1)(b-1) \geq 1$ for $a, b \geq e$).
Step 2. $\sqrt{x}$, $1/x$, e^{-x} are sub-additive or decreasing and violate the inequality (e.g. $\sqrt{x_1 + x_2} < \sqrt{x_1} + \sqrt{x_2}$).

GATE 2020 | GA Q.6 | 2 marks

The global financial crisis in 2008 is considered to be the most serious world-wide financial crisis, which started with the sub-prime lending crisis in USA in 2007. The sub-prime crisis led to the banking crisis in 2008 with the collapse of Lehman Brothers in 2008. The sub-prime lending refers to the provision of loans to those borrowers who may have difficulties in repaying loans, and it arises because of excess liquidity following the East Asian crisis. Which one of the following sequences shows the correct precedence as per the given passage?

(A) East Asian crisis → subprime lending crisis → banking crisis → global financial crisis.
(B) Subprime lending crisis → global financial crisis → banking crisis → East Asian crisis.
(C) Banking crisis → subprime lending crisis → global financial crisis → East Asian crisis.
(D) Global financial crisis → East Asian crisis → banking crisis → subprime lending crisis.

Answer: (A) [answer matches official key]

Solution. Step 1. Sub-prime lending arose from excess liquidity *following* the East Asian crisis, so East Asia comes first.
Step 2. The sub-prime crisis then led to the banking crisis, which became the global financial crisis.

GATE 2020 | GA Q.7 | 2 marks

It is quarter past three in your watch. The angle between the hour hand and the minute hand is _____.

(A) $0°$
(B) $7.5°$
(C) $15°$
(D) $22.5°$

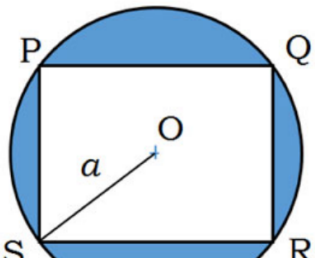
Answer: (B) [answer matches official key]

Solution. Step 1. At 3:15 the minute hand is at $15 \times 6° = 90°$.
Step 2. The hour hand is at $3 \times 30° + 15 \times 0.5° = 97.5°$.
Step 3. Angle $= 97.5 - 90 = 7.5°$.

GATE 2020 | GA Q.8 | 2 marks

A circle with centre O is shown in the figure. A rectangle $PQRS$ of maximum possible area is inscribed in the circle. If the radius of the circle is a, then the area of the shaded portion is _____.

is _____.



(A) $\pi a^2 - a^2$
(B) $\pi a^2 - \sqrt{2}a^2$
(C) $\pi a^2 - 2a^2$
(D) $\pi a^2 - 3a^2$

Answer: (C) [answer matches official key]

Solution. Step 1. The rectangle of maximum area inscribed in a circle is a square with diagonal $2a$.
Step 2. Side $= \sqrt{2}a$, area $= 2a^2$.
Step 3. Shaded area $= \pi a^2 - 2a^2$.

GATE 2020 | GA Q.9 | 2 marks

a, b, c are real numbers. The quadratic equation $ax^2 - bx + c = 0$ has equal roots, which is β, then

(A) $\beta = b/a$
(B) $\beta^2 = ac$
(C) $\beta^3 = bc/(2a^2)$
(D) $b \neq 4ac$

Answer: (C) [answer matches official key]

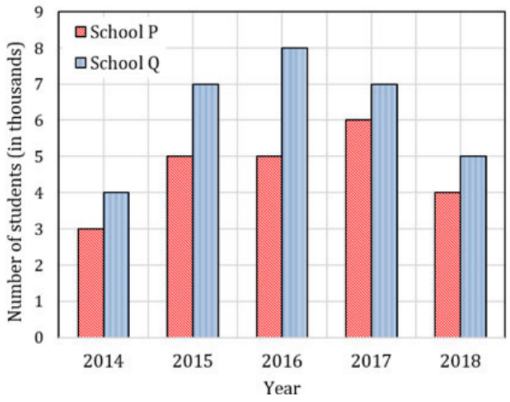
Solution. Step 1. Equal roots: $\beta = \dfrac{b}{2a}$ and $\beta^2 = \dfrac{c}{a}$ (sum and product of roots).

Step 2. $\beta^3 = \beta \cdot \beta^2 = \dfrac{b}{2a} \cdot \dfrac{c}{a} = \dfrac{bc}{2a^2}$.

GATE 2020 | GA Q.10 | 2 marks

The following figure shows the data of students enrolled in 5 years (2014 to 2018) for two schools P and Q. During this period, the ratio of the average number of the students enrolled in school P to the average of the difference of the number of students enrolled in schools P and Q is _____.

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(A) 8:23
(B) 23:8
(C) 23:31
(D) 31:23

Answer: (B) [answer matches official key]

Solution. Step 1. Read the chart (thousands): $P = 3, 5, 5, 6, 4$ (sum 23); $Q = 4, 7, 8, 7, 5$.
Step 2. Differences $|Q - P| = 1, 2, 3, 1, 1$ (sum 8).
Step 3. Ratio of averages $= \dfrac{23/5}{8/5} = 23 : 8$.

GATE 2020 | Q.1 | 1 mark

If $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_6$ are six vectors in $\mathbb{R}^4$, which one of the following statements is FALSE?

(A) It is not necessary that these vectors span $\mathbb{R}^4$.
(B) These vectors are not linearly independent.
(C) Any four of these vectors form a basis for $\mathbb{R}^4$.
(D) If $\{\mathbf{v}_1, \mathbf{v}_3, \mathbf{v}_5, \mathbf{v}_6\}$ spans $\mathbb{R}^4$, then it forms a basis for $\mathbb{R}^4$.

Answer: (C) [answer matches official key]

Solution. Step 1. Six vectors in $\mathbb{R}^4$ are always dependent (B true) and may fail to span (A true).
Step 2. Four vectors that span $\mathbb{R}^4$ are automatically a basis (D true).
Step 3. But *any* four of them need not be independent, so (C) is false.

GATE 2020 | Q.2 | 1 mark

For a vector field $\vec{A}$, which one of the following is FALSE?

(A) $\vec{A}$ is solenoidal if $\nabla \cdot \vec{A} = 0$.
(B) $\nabla \times \vec{A}$ is another vector field.
(C) $\vec{A}$ is irrotational if $\nabla^2 \vec{A} = 0$.
(D) $\nabla \times (\nabla \times \vec{A}) = \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A}$

Answer: (C) [answer matches official key]

Solution. Step 1. (A) is the definition of a solenoidal field, (B) is correct, and (D) is the standard vector identity.
Step 2. A field is irrotational iff $\nabla \times \vec{A} = 0$; $\nabla^2 \vec{A} = 0$ is not equivalent to that. So (C) is false.

GATE 2020 | Q.3 | 1 mark

The partial derivative of the function $f(x, y, z) = e^{1-x\cos y} + xze^{-1/(1+y^2)}$ with respect to x at the point $(1, 0, e)$ is

(A) -1
(B) 0
(C) 1
(D) $\dfrac{1}{e}$

Answer: (B) [answer matches official key]

Solution. Step 1. $\dfrac{\partial f}{\partial x} = -\cos y\, e^{1-x\cos y} + z\, e^{-1/(1+y^2)}$.
Step 2. At $(1, 0, e)$: $-1 \cdot e^{0} + e \cdot e^{-1} = -1 + 1 = 0$.

GATE 2020 | Q.4 | 1 mark

The general solution of $\dfrac{d^2y}{dx^2} - 6\dfrac{dy}{dx} + 9y = 0$ is

(A) $y = C_1e^{3x} + C_2e^{-3x}$
(B) $y = (C_1 + C_2x)e^{-3x}$
(C) $y = (C_1 + C_2x)e^{3x}$
(D) $y = C_1e^{3x}$

Answer: (C) [answer matches official key]

Solution. Step 1. Characteristic equation $\lambda^2 - 6\lambda + 9 = (\lambda - 3)^2 = 0$: repeated root 3.
Step 2. $y = (C_1 + C_2x)e^{3x}$.

GATE 2020 | Q.5 | 1 mark

The output $y[n]$ of a discrete-time system for an input $x[n]$ is $y[n] = \max_{-\infty \le k \le n} |x[k]|$. The unit impulse response of the system is

(A) 0 for all n.
(B) 1 for all n.
(C) unit step signal $u[n]$.
(D) unit impulse signal $\delta[n]$.

Answer: (C) [answer matches official key]

Solution. Step 1. For $x[n] = \delta[n]$: $y[n] = 0$ for $n < 0$ (the maximum over earlier samples is 0).
Step 2. For $n \ge 0$ the maximum includes $x[0] = 1$, so $y[n] = 1$.
Step 3. $h[n] = u[n]$.

GATE 2020 | Q.6 | 1 mark

A single crystal intrinsic semiconductor is at a temperature of 300 K with effective density of states for holes twice that of electrons. The thermal voltage is 26 mV. The intrinsic Fermi level is shifted from mid-bandgap energy level by

(A) 18.02 meV.
(B) 9.01 meV.
(C) 13.45 meV.
(D) 26.90 meV.

Answer: (B) [answer matches official key]

Solution. Step 1. $E_i - E_{mid} = \dfrac{kT}{2} \ln \dfrac{N_v}{N_c}$.

Step 2. $= \dfrac{26}{2} \ln 2 = 13 \times 0.693 = 9.01$ meV (towards the conduction band since $N_v > N_c$).

GATE 2020 | Q.7 | 1 mark

Consider the recombination process via bulk traps in a forward biased pn homojunction diode. The maximum recombination rate is U_{max}. If the electron and the hole capture cross-sections are equal, which one of the following is FALSE?

(A) With all other parameters unchanged, U_{max} decreases if the intrinsic carrier density is reduced.
(B) U_{max} occurs at the edges of the depletion region in the device.
(C) U_{max} depends exponentially on the applied bias.
(D) With all other parameters unchanged, U_{max} increases if the thermal velocity of the carriers increases.

Answer: (B) [answer matches official key]

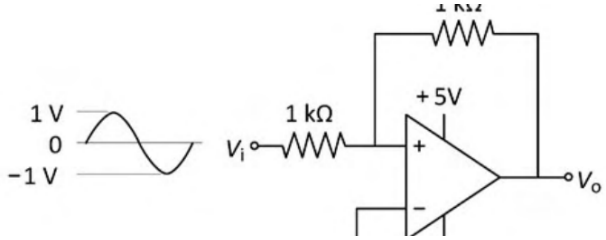
Solution. Step 1. In the SRH model, $U = \dfrac{\sigma v_{th} N_t \left(pn - n_i^2\right)}{n + p + 2n_i}$ (equal capture cross-sections).

Step 2. U is largest where $n \approx p$, which is near the *middle* of the depletion region, not at its edges. So (B) is false.

Step 3. (A), (D): $U \propto n_i$ and $\propto v_{th}$ at the maximum; (C): $U_{max} \propto e^{qV/2kT}$ depends exponentially on bias.

GATE 2020 | Q.8 | 1 mark

The components in the circuit shown below are ideal. If the op-amp is in positive feedback and the input voltage V_i is a sine wave of amplitude 1 V, the output voltage V_o is



(A) a non-inverting sine wave of 2 V amplitude.
(B) an inverted sine wave of 1 V amplitude.
(C) a square wave of 5 V amplitude.
(D) a constant of either +5 V or −5 V.

Answer: (D) [answer matches official key]

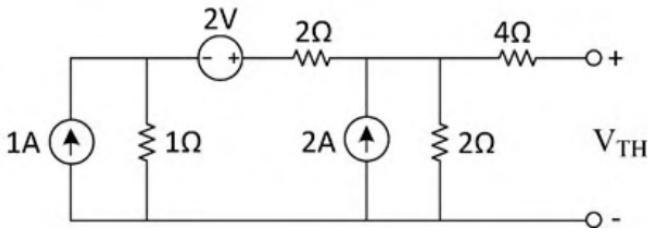
Solution. Step 1. With positive feedback the op-amp is a comparator; the output is ±5 V.

Step 2. The non-inverting input is $V_+ = \dfrac{V_i + V_o}{2}$ (two equal 1 kΩ resistors) while the inverting input is at 0.

Step 3. If $V_o = +5$: $V_+ = \dfrac{V_i + 5}{2} > 0$ for all $|V_i| \le 1$, so the output stays at +5 V; likewise −5 V stays. The circuit latches: V_o is a constant, either +5 V or −5 V.

GATE 2020 | Q.9 | 1 mark

In the circuit shown below, the Thevenin voltage V_{TH} is



(A) 2.4 V
(B) 2.8 V
(C) 3.6 V
(D) 4.5 V

Answer: (C) [answer matches official key]

Solution. Step 1. No current flows in the 4 Ω resistor (open terminals), so V_{TH} is the voltage of the right-hand node V_2.

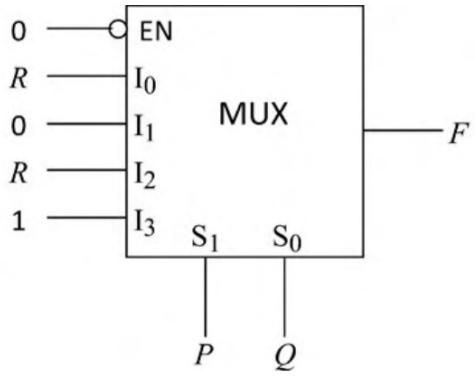
Step 2. Let V_1 be the left node voltage and $i = \dfrac{V_1 + 2 - V_2}{2}$ the current in the top 2 Ω branch (with the 2 V source polarity as drawn).

Step 3. Left node: $1 = \dfrac{V_1}{1} + i \Rightarrow V_1 = 1 - i$. Right node: $2 + i = \dfrac{V_2}{2} \Rightarrow V_2 = 4 + 2i$.

Step 4. $2i = 1 - i + 2 - 4 - 2i \Rightarrow 5i = -1 \Rightarrow i = -0.2$ A, so $V_{TH} = V_2 = 4 - 0.4 = 3.6$ V.

GATE 2020 | Q.10 | 1 mark

The figure below shows a multiplexer where S_1 and S_0 are the select lines, I_0 to I_3 are the input data lines, EN is the enable line, and $F(P, Q, R)$ is the output. F is



(A) $PQ + \bar{Q}R$.

(A) $PQ + \overline{Q}R$
(B) $P + Q\overline{R}$
(C) $P\overline{Q}R + \overline{P}Q$
(D) $\overline{Q} + PR$

Answer: (A) [answer matches official key]

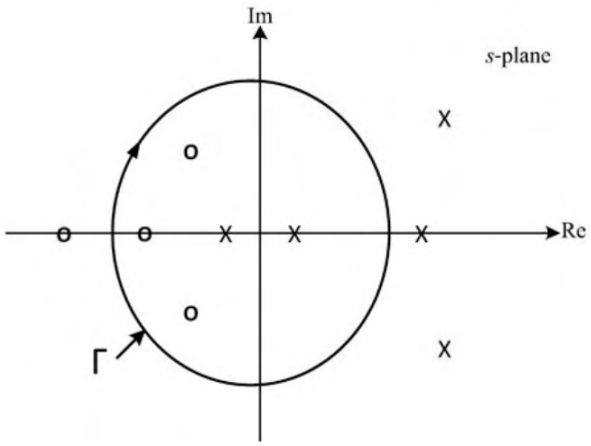
Solution. Step 1. The enable is tied active ($EN = 0$ for an active-low enable). Select lines: $S_1 = P$, $S_0 = Q$. Data inputs: $I_0 = R$, $I_1 = 0$, $I_2 = R$, $I_3 = 1$.

Step 2. $F = \overline{P}\,\overline{Q}\,R + \overline{P}Q \cdot 0 + P\overline{Q}R + PQ \cdot 1 = \overline{Q}R(\overline{P} + P) + PQ$.

Step 3. $F = \overline{Q}R + PQ$.

GATE 2020 | Q.11 | 1 mark

The pole-zero map of a rational function $G(s)$ is shown below. When the closed contour Γ is mapped into the $G(s)$-plane, then the mapping encircles



(A) the origin of the $G(s)$-plane once in the counter-clockwise direction.

(A) the origin of the $G(s)$-plane once in the counter-clockwise direction.
(B) the origin of the $G(s)$-plane once in the clockwise direction.
(C) the point $-1 + j0$ of the $G(s)$-plane once in the counter-clockwise direction.
(D) the point $-1 + j0$ of the $G(s)$-plane once in the clockwise direction.

Answer: (B) [answer matches official key]

Solution. Step 1. The contour Γ (clockwise) encloses Z zeros and P poles of $G(s)$; the image of Γ encircles the *origin* $N = Z - P$ times in the clockwise sense (argument principle).
Step 2. Counting the circles ($\circ$, zeros) and crosses ($\times$, poles) inside Γ gives $Z - P = 1$: one clockwise encirclement of the origin.

GATE 2020 | Q.12 | 1 mark

A digital communication system transmits a block of N bits. The probability of error in decoding a bit is α. The error event of each bit is independent of the error events of the other bits. The received block is declared erroneous if at least one of its bits is decoded wrongly. The probability that the received block is erroneous is

(A) $N(1-\alpha)$
(B) α^N
(C) $1-\alpha^N$
(D) $1-(1-\alpha)^N$

Answer: (D) [answer matches official key]

Solution. Step 1. The block is correct only if all N bits are correct: probability $(1-\alpha)^N$.
Step 2. $P(\text{block error}) = 1-(1-\alpha)^N$.

GATE 2020 | Q.13 | 1 mark

The impedances $Z = jX$, for all X in the range $(-\infty, \infty)$, map to the Smith chart as

(A) a circle of radius 1 with centre at $(0, 0)$.
(B) a point at the centre of the chart.
(C) a line passing through the centre of the chart.
(D) a circle of radius 0.5 with centre at $(0.5, 0)$.

Answer: (A) [answer matches official key]

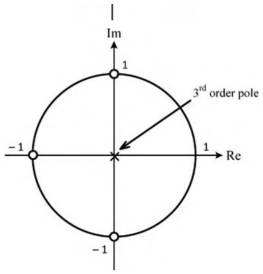
Solution. Step 1. A purely reactive impedance has $\mathrm{Re}\{z\} = 0$, giving $|\Gamma| = \left|\dfrac{jx-1}{jx+1}\right| = 1$ for all x.

Step 2. Hence it maps onto the outer boundary circle of unit radius centred at the origin.

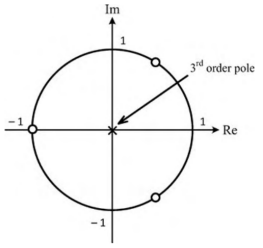
GATE 2020 | Q.14 | 1 mark

Which one of the following pole-zero plots corresponds to the transfer function of an LTI system characterized by the input-output difference equation given below? $y[n] = \sum_{k=0}^{3}(-1)^k x[n-k]$

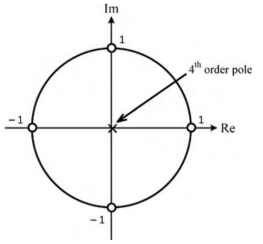
(B)



(C)



(D)



Answer: (A) [answer matches official key]

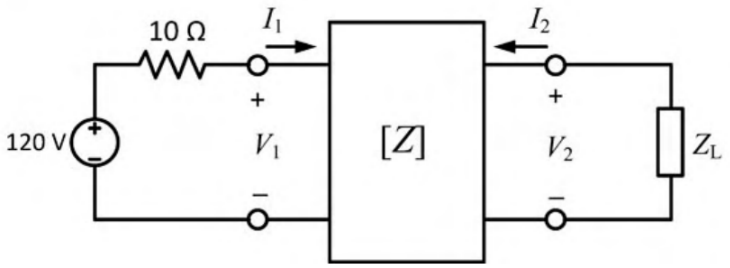
Solution. Step 1. $H(z) = 1 - z^{-1} + z^{-2} - z^{-3} = \dfrac{1-z^{-4}}{1+z^{-1}} = \dfrac{z^4-1}{z^3(z+1)}$.

Step 2. Zeros: the fourth roots of unity except $z = -1$ (cancelled): $z = 1,\ j,\ -j$. Poles: a third-order pole at $z = 0$.

Step 3. This is plot (A): three zeros on the unit circle at $1, \pm j$ and a 3rd-order pole at the origin.

GATE 2020 | Q.15 | 1 mark

In the given circuit, the two-port network has the impedance matrix $[Z] = \begin{bmatrix} 40 & 60 \\ 60 & 120 \end{bmatrix}\ \Omega$. The value of Z_L for which maximum power is transferred to the load is _____ Ω.



Answer: 48 [answer matches official key]

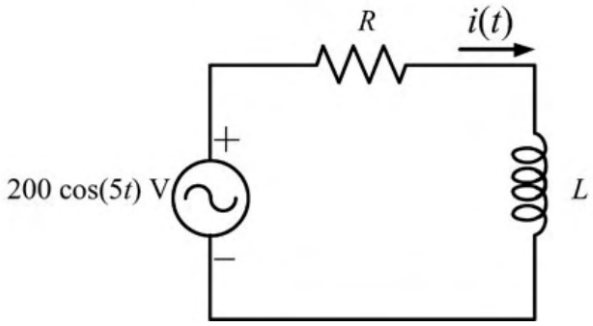
Solution. Step 1. Maximum power: $Z_L = Z_{th}^*$, the output impedance seen at port 2 with the source shorted.

Step 2. The source side has 10 Ω: $V_1 = -10I_1$ (source off). From $V_1 = 40I_1 + 60I_2$: $I_1 = -\dfrac{60}{50}I_2$.

Step 3. $V_2 = 60I_1 + 120I_2 = \left(120 - \dfrac{3600}{50}\right) I_2 = 48\, I_2$, so $Z_{th} = 48\ \Omega$ and $Z_L = 48\ \Omega$.

GATE 2020 | Q.16 | 1 mark

The current in the RL-circuit shown below is $i(t) = 10\cos(5t - \pi/4)$ A. The value of the inductor (rounded off to two decimal places) is _____ H.



Answer: 2.83 [answer matches official key]

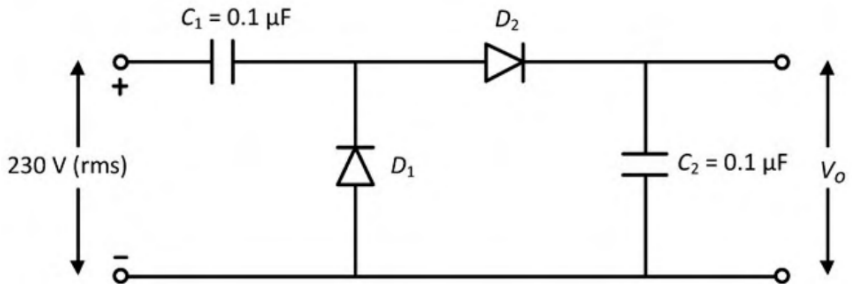
Solution. Step 1. $Z = \dfrac{200\angle 0^\circ}{10\angle -45^\circ} = 20\angle 45^\circ = 14.14 + j14.14\ \Omega$.

Step 2. $\omega L = 14.14 \Rightarrow L = \dfrac{14.14}{5} = 2.83$ H.

GATE 2020 | Q.17 | 1 mark

In the circuit shown below, all the components are ideal and the input voltage is sinusoidal. The magnitude of the steady-state output V_o (rounded off to two decimal places) is _____ V.

places) is _____ V.

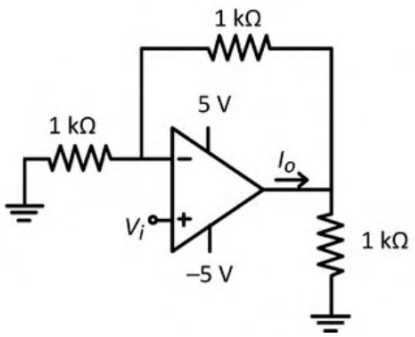


Answer: 650.54 [answer matches official key]

Solution. Step 1. This is a voltage doubler (clamp C_1, D_1 and peak detector D_2, C_2).
Step 2. C_1 charges to the peak $V_m = 230\sqrt{2} = 325.27$ V through D_1.
Step 3. C_2 then charges to $V_m + V_m = 2V_m = 650.54$ V.

GATE 2020 | Q.18 | 1 mark

In the circuit shown below, all the components are ideal. If V_I is +2 V, the current I_o sourced by the op-amp is _____ mA.



Answer: 6 [answer matches official key]

Solution. Step 1. The amplifier is non-inverting with $R_f = R_g = 1$ kΩ: $V_o = (1+1) \times 2 = 4$ V (within the ±5 V supplies).
Step 2. Load current $= \dfrac{4}{1\ \text{k}\Omega} = 4$ mA; feedback current $= \dfrac{4}{2\ \text{k}\Omega} = 2$ mA.
Step 3. $I_o = 4 + 2 = 6$ mA.

GATE 2020 | Q.19 | 1 mark

In an 8085 microprocessor, the number of address lines required to access a 16 K byte memory bank is _____.

Answer: 14 [answer matches official key]

Solution. Step 1. 16 K $= 2^{14}$ locations, so 14 address lines are needed.

GATE 2020 | Q.20 | 1 mark

A 10-bit D/A converter is calibrated over the full range from 0 to 10 V. If the input to the D/A converter is $13A$ (in hex), the output (rounded off to three decimal places) is _____ V.

Answer: 3.069 [answer matches official key]

Solution. Step 1. $13A_{16} = 1 \cdot 256 + 3 \cdot 16 + 10 = 314$.
Step 2. Full scale = 1023 steps for 10 V: $V_o = \dfrac{314}{1023} \times 10 = 3.069$ V.

GATE 2020 | Q.21 | 1 mark

A transmission line of length $3\lambda/4$ and having a characteristic impedance of 50 Ω is terminated with a load of 400 Ω. The impedance (rounded off to two decimal places) seen at the input end of the transmission line is _____ Ω.

Answer: 6.25 [answer matches official key]

Solution. Step 1. An odd multiple of a quarter wavelength acts as an impedance inverter: $Z_{in} = \dfrac{Z_0^2}{Z_L}$.

Step 2. $Z_{in} = \dfrac{50^2}{400} = 6.25\ \Omega$.

GATE 2020 | Q.22 | 1 mark

A binary random variable X takes the value $+2$ or -2. The probability $P(X = +2) = \alpha$. The value of α (rounded off to one decimal place), for which the entropy of X is maximum, is _____.

Answer: 0.5 [answer matches official key]

Solution. Step 1. $H = -\alpha \log_2 \alpha - (1-\alpha)\log_2(1-\alpha)$ is maximum when the two outcomes are equally likely.
Step 2. $\alpha = 0.5$.

GATE 2020 | Q.23 | 1 mark

The loop transfer function of a negative feedback system is $G(s)H(s) = \dfrac{K(s+11)}{s(s+2)(s+8)}$. The value of K, for which the system is marginally stable, is _____.

Answer: 160 [answer matches official key]

Solution. Step 1. Characteristic equation: $s^3 + 10s^2 + 16s + K(s+11) = s^3 + 10s^2 + (16+K)s + 11K = 0$.
Step 2. Routh: the s^1 row vanishes when $\dfrac{10(16+K) - 11K}{10} = 0 \Rightarrow 160 - K = 0$.
Step 3. $K = 160$.

GATE 2020 | Q.24 | 1 mark

The random variable $Y = \displaystyle\int_{-\infty}^{\infty} W(t)\phi(t)\,dt$, where $\phi(t) = \begin{cases} 1, & 5 \le t \le 7 \\ 0, & \text{otherwise} \end{cases}$ and $W(t)$ is a real white Gaussian noise process with two-sided power spectral density $S_W(f) = 3$ W/Hz, for all f. The variance of Y is _____.

Answer: 6 [answer matches official key]

Solution. Step 1. For white noise, $\text{Var}(Y) = S_W \int |\phi(t)|^2 dt$.
Step 2. $\int \phi^2 dt = 2$, so $\text{Var}(Y) = 3 \times 2 = 6$.

GATE 2020 | Q.25 | 1 mark

The two sides of a *fair* coin are labelled as 0 and 1. The coin is tossed two times independently. Let M and N denote the labels corresponding to the outcomes of those tosses. For a random variable X, defined as $X = \min(M, N)$, the expected value $E(X)$ (rounded off to two decimal places) is _____.

Answer: 0.25 [answer matches official key]

Solution. Step 1. $X = 1$ only if both tosses give 1: $P = \frac{1}{4}$; otherwise $X = 0$.
Step 2. $E[X] = 0.25$.

GATE 2020 | Q.26 | 2 marks

Consider the following system of linear equations. $x_1 + 2x_2 = b_1$; $2x_1 + 4x_2 = b_2$; $3x_1 + 7x_2 = b_3$; $3x_1 + 9x_2 = b_4$. Which one of the following conditions ensures that a solution exists for the above system?

(A) $b_2 = 2b_1$ and $6b_1 - 3b_3 + b_4 = 0$
(B) $b_3 = 2b_1$ and $6b_1 - 3b_3 + b_4 = 0$
(C) $b_2 = 2b_1$ and $3b_1 - 6b_3 + b_4 = 0$

(D) $b_3 = 2b_1$ and $3b_1 - 6b_3 + b_4 = 0$

Answer: (A) [answer matches official key]

Solution. Step 1. Equations 1 and 2 are proportional only if $b_2 = 2b_1$.
Step 2. Solve equations 1 and 3: $x_2 = b_3 - 3b_1$, $x_1 = b_1 - 2x_2 = 7b_1 - 2b_3$.
Step 3. Substitute in equation 4: $3(7b_1 - 2b_3) + 9(b_3 - 3b_1) = -6b_1 + 3b_3 = b_4 \Rightarrow 6b_1 - 3b_3 + b_4 = 0$.

GATE 2020 | Q.27 | 2 marks

Which one of the following options contains two solutions of the differential equation $\dfrac{dy}{dx} = (y-1)x$?

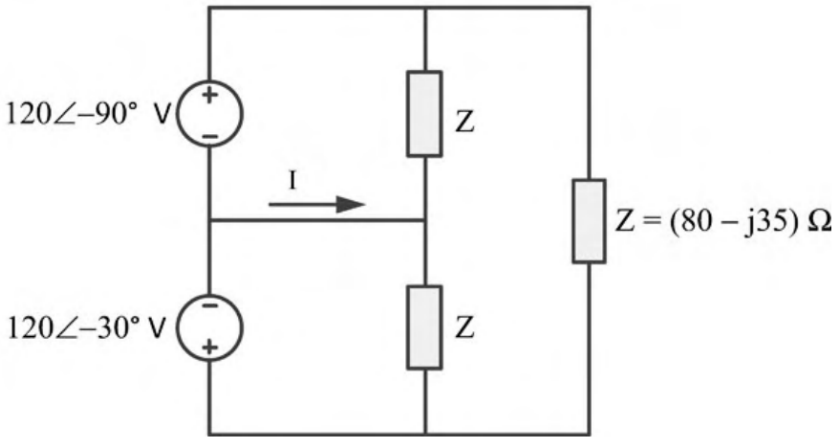
(A) $\ln|y-1| = 0.5x^2 + C$ and $y = 1$
(B) $\ln|y-1| = 2x^2 + C$ and $y = 1$
(C) $\ln|y-1| = 0.5x^2 + C$ and $y = -1$
(D) $\ln|y-1| = 2x^2 + C$ and $y = -1$

Answer: (A) [answer matches official key]

Solution. Step 1. Separable: $\dfrac{dy}{y-1} = x\,dx \Rightarrow \ln|y-1| = \dfrac{x^2}{2} + C$.
Step 2. The constant solution $y = 1$ (which makes the right side zero) is also a solution.

GATE 2020 | Q.28 | 2 marks

The current I in the given network is



(A) 0 A.
(B) $2.38\angle -96.37^\circ$ A.
(C) $2.38\angle 143.63^\circ$ A.
(D) $2.38\angle -23.63^\circ$ A.

Answer: (C) [answer matches official key]

Solution. Step 1. Take the middle wire as reference. The two series impedances Z and the load $Z_L = 80 - j35\ \Omega$ (equal values Z) meet at the right node with voltage V_R.
Step 2. KCL at V_R: $\dfrac{V_1 - V_R}{Z} + \dfrac{V_2 - V_R}{Z} = \dfrac{V_R}{Z} \Rightarrow V_R = \dfrac{V_1 + V_2}{3}$, where $V_1 = 120\angle -90^\circ$ and $V_2 = 120\angle -30^\circ$ (as drawn).
Step 3. $V_1 + V_2 = -j120 + (103.9 - j60) = 103.9 - j180 = 207.8\angle -60^\circ$.
Step 4. $|I| = \dfrac{|V_1 + V_2|}{|Z|} = \dfrac{207.8}{87.3} = 2.38$ A; with the direction of I as drawn (opposite to the sum) the angle is $-60^\circ + 23.63^\circ + 180^\circ = 143.63^\circ$.

GATE 2020 | Q.29 | 2 marks

A finite duration discrete-time signal $x[n]$ is obtained by sampling the continuous-time signal $x(t) = \cos(200\pi t)$ at sampling instants $t = n/400$, $n = 0, 1, \ldots, 7$. The 8-point discrete Fourier transform (DFT) of $x[n]$ is defined as $X[k] = \sum_{n=0}^{7} x[n] e^{-j\frac{\pi kn}{4}}$, $k = 0, 1, \ldots, 7$. Which one of the following statements is TRUE?

(A) All $X[k]$ are non-zero.
(B) Only $X[4]$ is non-zero.
(C) Only $X[2]$ and $X[6]$ are non-zero.
(D) Only $X[3]$ and $X[5]$ are non-zero.

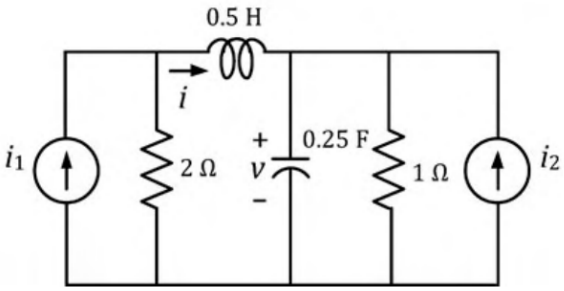
Answer: (C) [answer matches official key]

Solution. Step 1. $x[n] = \cos\left(200\pi \frac{n}{400}\right) = \cos\frac{\pi n}{2} = \cos\frac{2\pi \cdot 2n}{8}$.

Step 2. A cosine with exactly 2 cycles in the 8 samples has non-zero DFT only at $k = 2$ and $k = 8 - 2 = 6$.

GATE 2020 | Q.30 | 2 marks

For the given circuit, which one of the following is the correct state equation?



(A) $\frac{d}{dt}\begin{bmatrix} v \\ i \end{bmatrix} = \begin{bmatrix} -4 & 4 \\ -2 & -4 \end{bmatrix}\begin{bmatrix} v \\ i \end{bmatrix} + \begin{bmatrix} 0 & 4 \\ 4 & 0 \end{bmatrix}\begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$

(B) $\frac{d}{dt}\begin{bmatrix} v \\ i \end{bmatrix} = \begin{bmatrix} -4 & -4 \\ -2 & 4 \end{bmatrix}\begin{bmatrix} v \\ i \end{bmatrix} + \begin{bmatrix} 4 & 4 \\ 4 & 0 \end{bmatrix}\begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$

(C) $\frac{d}{dt}\begin{bmatrix} v \\ i \end{bmatrix} = \begin{bmatrix} 4 & -4 \\ -2 & -4 \end{bmatrix}\begin{bmatrix} v \\ i \end{bmatrix} + \begin{bmatrix} 0 & 4 \\ 4 & 4 \end{bmatrix}\begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$

(D) $\frac{d}{dt}\begin{bmatrix} v \\ i \end{bmatrix} = \begin{bmatrix} -4 & -4 \\ -2 & -4 \end{bmatrix}\begin{bmatrix} v \\ i \end{bmatrix} + \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}\begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$

Answer: (A) [answer matches official key]

Solution. Step 1. Take the capacitor voltage v and inductor current i as states.

Step 2. Capacitor node: $0.25\frac{dv}{dt} = i - \frac{v}{1} + i_2 \Rightarrow \frac{dv}{dt} = -4v + 4i + 4i_2$.

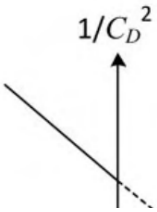
Step 3. Inductor: $0.5\frac{di}{dt} = v_A - v$ with the left node voltage $v_A = 2(i_1 - i)$, so $\frac{di}{dt} = -2v - 4i + 4i_1$.

Step 4. In matrix form this is option (A).

GATE 2020 | Q.31 | 2 marks

A one-sided abrupt pn junction diode has a depletion capacitance C_D of 50 pF at a reverse bias of 0.2 V. The plot of $1/C_D^2$ versus the applied voltage V for this diode is a straight line as shown in the figure below. The slope of the plot is _____ $\times 10^{20}$ $\mathrm{F}^{-2}\mathrm{V}^{-1}$.

is _____ $\times 10^{20}\ F^{-2}V^{-1}$.



(A) -5.7
(B) -3.8
(C) -1.2
(D) -0.4

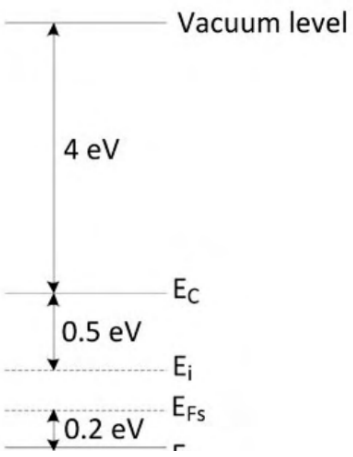
Answer: Marks to all [answer matches official key]

Solution. Step 1. For a one-sided abrupt junction $\frac{1}{C_D^2} = \frac{2(V_{bi} - V)}{q\varepsilon N A^2}$, a straight line with negative slope in V.

Step 2. The slope cannot be determined from the data given (C_D at one bias only, no V_{bi} or doping/area), so none of the options can be derived. The official answer key awards marks to all candidates for this question.

GATE 2020 | Q.32 | 2 marks

The band diagram of a p-type semiconductor with a band-gap of 1 eV is shown. Using this semiconductor, a MOS capacitor having V_{TH} of -0.16 V, C'_{ox} of 100 nF/cm^2 and a metal work function of 3.87 eV is fabricated. There is no charge within the oxide. If the voltage across the capacitor is V_{TH}, the magnitude of depletion charge per unit area (in C/cm^2) is



(A) 1.70×10^{-8}
(B) 0.52×10^{-8}
(C) 1.41×10^{-8}
(D) 0.93×10^{-8}

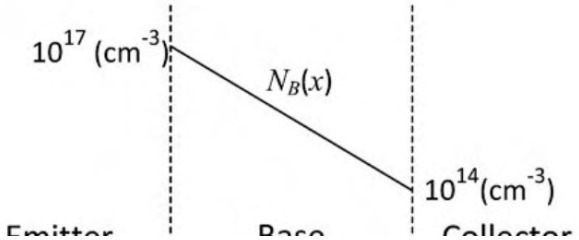
Answer: (A) [answer matches official key]

Solution. Step 1. From the diagram $\chi = 4$ eV, $E_C - E_F = 0.8$ eV, so $\phi_s = 4.8$ eV and $\phi_{ms} = 3.87 - 4.8 = -0.93$ V.
Step 2. E_F lies 0.3 eV below E_i: $\phi_F = 0.3$ V, so $2\phi_F = 0.6$ V.
Step 3. $V_{TH} = \phi_{ms} + 2\phi_F + \frac{|Q_d|}{C_{ox}} \Rightarrow -0.16 = -0.93 + 0.6 + \frac{|Q_d|}{C_{ox}}$, giving $\frac{|Q_d|}{C_{ox}} = 0.17$ V.
Step 4. $|Q_d| = 0.17 \times 100$ nF/cm^2 $= 1.70 \times 10^{-8}$ C/cm^2.

GATE 2020 | Q.33 | 2 marks

The base of an *npn* BJT T_1 has a linear doping profile $N_B(x)$ as shown below. The base of another *npn* BJT T_2 has a uniform doping N_B of 10^{17} cm^{-3}. All other parameters are identical for both the devices. Assuming that the hole density profile is the same as that of doping, the common-emitter current gain of T_2 is

profile is the same as that of doping, the common-emitter current gain of T2 is



(A) approximately 2.0 times that of T_1.
(B) approximately 0.3 times that of T_1.
(C) approximately 2.5 times that of T_1.
(D) approximately 0.7 times that of T_1.

Answer: Marks to all [answer matches official key]

Solution. Step 1. The current gain $\beta \propto \dfrac{\int_0^W N_B\,dx}{D_n \ldots}$ (Gummel number), so one compares $\int N_B\,dx$ for the two bases: T_1 has an average doping near 5×10^{16} and T_2 has 10^{17}.
Step 2. The ratio depends on how the Gummel number and diffusion coefficients are treated, and the question was found ambiguous: the official key awards marks to all.

GATE 2020 | Q.34 | 2 marks

A *pn* junction solar cell of area 1.0 cm^2, illuminated uniformly with 100 mW cm^{-2}, has the following parameters: Efficiency = 15%, open circuit voltage = 0.7 V, fill factor = 0.8, and thickness = 200 μm. The charge of an electron is 1.6×10^{-19} C. The average optical generation rate (in cm^{-3}s^{-1}) is

(A) 0.84×10^{19}.
(B) 5.57×10^{19}.
(C) 1.04×10^{19}.
(D) 83.60×10^{19}.

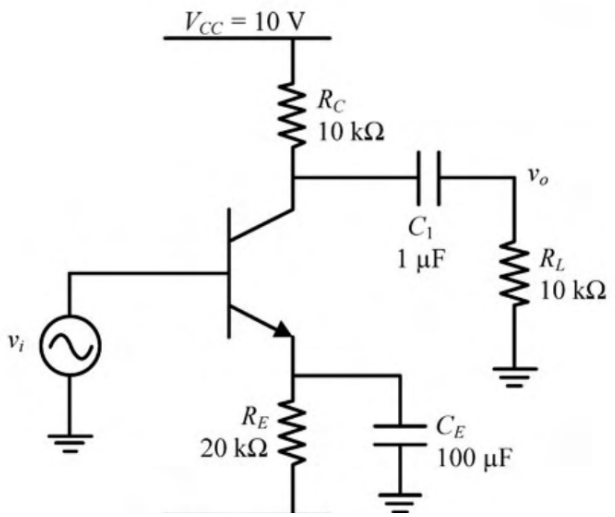
Answer: (A) [answer matches official key]

Solution. Step 1. Output power $P = 0.15 \times 100 = 15$ mW. Also $P = V_{oc} I_{sc}\, FF \Rightarrow I_{sc} = \dfrac{15 \text{ mW}}{0.7 \times 0.8} = 26.8$ mA.

Step 2. $G = \dfrac{I_{sc}}{q\,(\text{volume})} = \dfrac{26.8 \times 10^{-3}}{1.6 \times 10^{-19} \times 1 \times 200 \times 10^{-4}} = 8.4 \times 10^{18} \approx 0.84 \times 10^{19}$ cm^{-3}s^{-1}.

GATE 2020 | Q.35 | 2 marks

For the BJT in the amplifier shown below, $V_{BE} = 0.7$ V, $kT/q = 26$ mV. Assume that BJT output resistance (r_o) is very high and the base current is negligible. The capacitors are also assumed to be short circuited at signal frequencies. The input v_i is direct coupled. The low frequency voltage gain v_o/v_i of the amplifier is



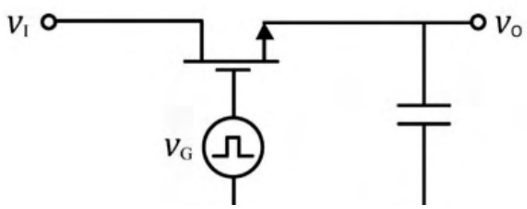
(A) -89.42
(B) -128.21
(C) -178.85
(D) -256.42

Answer: (A) [answer matches official key]

Solution. Step 1. DC: base at 0 V (direct-coupled input), so $I_E = \dfrac{0 - 0.7 - (-10)}{20\ \text{k}\Omega} = 0.465\ \text{mA} \approx I_C$.
Step 2. $g_m = \dfrac{I_C}{V_T} = \dfrac{0.465\ \text{mA}}{26\ \text{mV}} = 17.9\ \text{mS}$.
Step 3. C_E bypasses R_E at signal frequencies, so $A_v = -g_m(R_C \parallel R_L) = -17.9\ \text{mS} \times 5\ \text{k}\Omega = -89.4$.

GATE 2020 | Q.36 | 2 marks

An enhancement MOSFET of threshold voltage 3 V is being used in the sample and hold circuit given below. Assume that the substrate of the MOS device is connected to −10 V. If the input voltage V_I lies between ±10 V, the minimum and the maximum values of V_G required for proper sampling and holding respectively, are



(A) 3 V and −3 V.
(B) 10 V and −10 V.
(C) 13 V and −7 V.
(D) 10 V and −13 V.

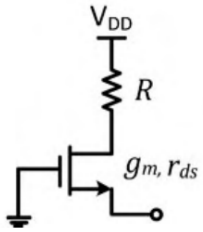
Answer: (C) [answer matches official key]

Solution. Step 1. Sampling: the transistor must conduct for the highest input: $V_G \geq V_{I,max} + V_T = 10 + 3 = 13$ V.
Step 2. Holding: it must be off for the lowest input, whose source can sit at −10 V: $V_G < V_S + V_T = -10 + 3 = -7$ V.
Step 3. So the gate pulse must swing between 13 V and −7 V.

GATE 2020 | Q.37 | 2 marks

Using the incremental low frequency small-signal model of the MOS device, the Norton equivalent resistance of the following circuit is

Norton equivalent resistance of the following circuit is



(A) $r_{ds} + R + g_m r_{ds} R$

(A) $r_{ds} + R + g_m r_{ds} R$

(B) $\dfrac{r_{ds} + R}{1 + g_m r_{ds}}$

(C) $r_{ds} + \dfrac{1}{g_m} + R$

(D) $r_{ds} + R$

Answer: (B) [answer matches official key]

Solution. Step 1. The gate is at ac ground and the terminal is the source, so this is a common-gate stage whose drain is loaded by R (the supply is an ac ground).

Step 2. Apply a test voltage v_t at the source and let i_d be the drain-to-source current. Then $v_{gs} = -v_t$, $v_d = -i_d R$ and $i_d = g_m v_{gs} + \dfrac{v_d - v_t}{r_{ds}}$.

Step 3. $i_d = -g_m v_t - \dfrac{i_d R + v_t}{r_{ds}} \Rightarrow i_d \left(1 + \dfrac{R}{r_{ds}}\right) = -v_t \left(g_m + \dfrac{1}{r_{ds}}\right)$.

Step 4. The test current into the source is $i_t = -i_d = v_t \dfrac{1 + g_m r_{ds}}{r_{ds} + R}$, hence $R_N = \dfrac{v_t}{i_t} = \dfrac{r_{ds} + R}{1 + g_m r_{ds}}$.

GATE 2020 | Q.38 | 2 marks

P, Q, and R are the decimal integers corresponding to the 4-bit binary number 1100 considered in signed magnitude, 1's complement, and 2's complement representations, respectively. The 6-bit 2's complement representation of $(P + Q + R)$ is

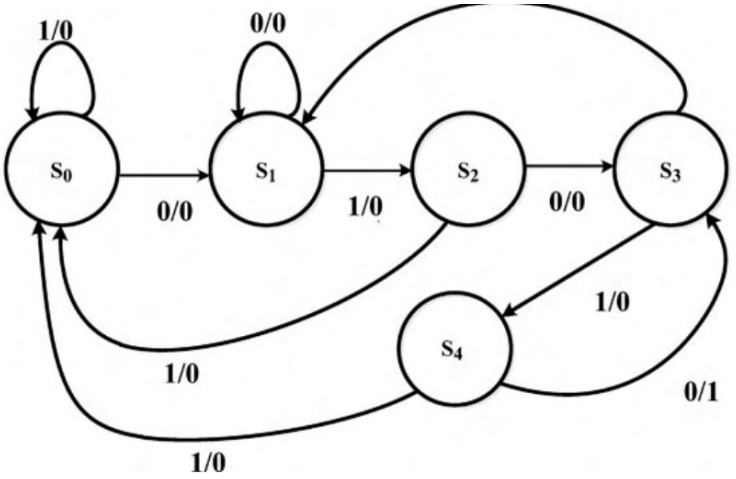
(A) 110101
(B) 110010
(C) 111101
(D) 111001

Answer: (A) [answer matches official key]

Solution. Step 1. Signed magnitude: sign bit 1, magnitude $100 = 4$: $P = -4$.
Step 2. 1's complement: invert $1100 \to 0011 = 3$: $Q = -3$.
Step 3. 2's complement: $1100 = -8 + 4$: $R = -4$.
Step 4. $P + Q + R = -11$; in 6 bits $64 - 11 = 53 = 110101_2$.

GATE 2020 | Q.39 | 2 marks

The state diagram of a sequence detector is shown below. State S_0 is the initial state of the sequence detector. If the output is 1, then



(A) the sequence 01010 is detected.

(A) the sequence 01010 is detected.
(B) the sequence 01011 is detected.
(C) the sequence 01110 is detected.
(D) the sequence 01001 is detected.

Answer: (A) [answer matches official key]

Solution. Step 1. The only transition with output 1 is the last one on the path $S_0 \xrightarrow{0} S_1 \xrightarrow{1} S_2 \xrightarrow{0} S_3 \xrightarrow{1} S_4 \xrightarrow{0/1}$.
Step 2. Reading the input labels along it gives $0, 1, 0, 1, 0$: the sequence 01010.

GATE 2020 | Q.40 | 2 marks

The characteristic equation of a system is $s^3 + 3s^2 + (K+2)s + 3K = 0$. In the root locus plot for the given system, as K varies from 0 to ∞, the break-away or break-in point(s) lie within

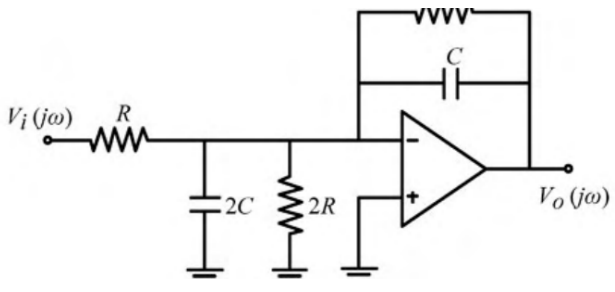
(A) $(-1, 0)$.
(B) $(-2, -1)$.
(C) $(-3, -2)$.
(D) $(-\infty, -3)$.

Answer: (A) [answer matches official key]

Solution. Step 1. $1 + K\dfrac{s+3}{s(s+1)(s+2)} = 0 \Rightarrow K = -\dfrac{s^3+3s^2+2s}{s+3}$.
Step 2. $\dfrac{dK}{ds} = 0 \Rightarrow s^3 + 6s^2 + 9s + 3 = 0$, whose roots are $-0.47,\ -1.65,\ -3.88$.
Step 3. The real-axis locus lies on $[-1, 0]$ and $[-3, -2]$; only $s = -0.47$ lies on it: the break-away point is in $(-1, 0)$.

GATE 2020 | Q.41 | 2 marks

The components in the circuit given below are ideal. If $R = 2$ kΩ and $C = 1$ μF, the -3 dB cut-off frequency of the circuit in Hz is



(A) 14.92
(B) 34.46
(C) 59.68
(D) 79.58

Answer: (D) [answer matches official key]

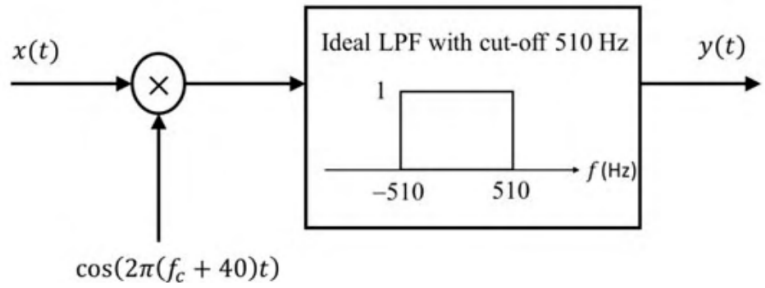
Solution. Step 1. The op-amp circuit has input resistor R and feedback network $R \parallel C$, giving a first-order low-pass response $\dfrac{V_o}{V_i} \propto \dfrac{1}{1+sRC}$.

Step 2. $f_c = \dfrac{1}{2\pi RC} = \dfrac{1}{2\pi \times 2 \times 10^3 \times 10^{-6}} = 79.58$ Hz.

GATE 2020 | Q.42 | 2 marks

For the modulated signal $x(t) = m(t)\cos(2\pi f_c t)$, the message signal $m(t) = 4\cos(1000\pi t)$ and the carrier frequency f_c is 1 MHz. The signal $x(t)$ is passed through a demodulator, as shown in the figure below. The output $y(t)$ of the demodulator is

For the modulated signal $x(t) = m(t)\cos(2\pi f_c t)$, the message signal $m(t) = 4\cos(1000\pi t)$ and the carrier frequency f_c is 1 MHz. The signal $x(t)$ is passed through a demodulator, as shown in the figure below. The output $y(t)$ of the demodulator is



$\cos(460\pi t)$.
$\cos(920\pi t)$.
$\cos(1000\pi t)$.
$\cos(540\pi t)$.

(A) $\cos(460\pi t)$.
(B) $\cos(920\pi t)$.
(C) $\cos(1000\pi t)$.
(D) $\cos(540\pi t)$.

Answer: (B) [answer matches official key]

Solution. Step 1. Multiply by $\cos(2\pi(f_c+40)t)$: $x(t)\cos(2\pi(f_c+40)t) = \dfrac{m(t)}{2}[\cos(2\pi\, 40\, t) + \cos(2\pi(2f_c+40)t)]$.

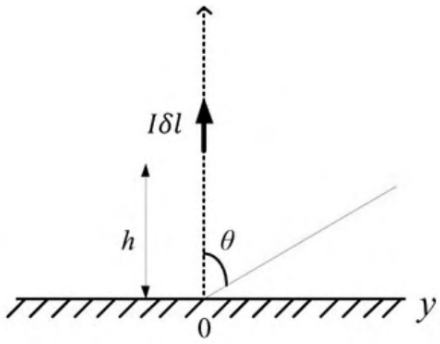
Step 2. The low-pass filter (510 Hz) keeps the first term: $2\cos(1000\pi t)\cos(80\pi t) = \cos(1080\pi t) + \cos(920\pi t)$.

Step 3. 1080π rad/s is 540 Hz $>$ 510 Hz and is rejected; 920π rad/s (460 Hz) passes: $y = \cos(920\pi t)$.

GATE 2020 | Q.43 | 2 marks

For an infinitesimally small dipole in free space, the electric field E_θ in the far field is proportional to

$\left(\frac{e^{-jkr}}{r}\right)\sin\theta$, where $k = 2\pi/\lambda$. A vertical infinitesimally small electric dipole ($\delta l \ll \lambda$) is placed at a distance h ($h > 0$) above an infinite ideal conducting plane, as shown in the figure. The minimum value of h, for which one of the maxima in the far field radiation pattern occurs at $\theta = 60°$, is



Infinite conducting plane

(A) λ
(B) 0.5λ
(C) 0.25λ
(D) 0.75λ

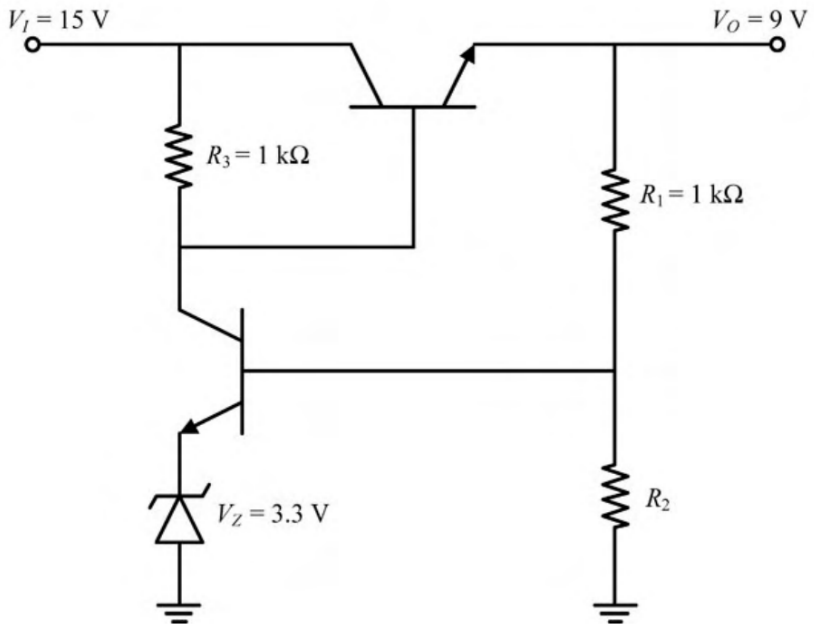
Answer: (A) [answer matches official key]

Solution. Step 1. By images, a vertical dipole above a perfect conductor has an image dipole of the same direction at distance $2h$: array factor $2\cos(kh\cos\theta)$.
Step 2. The total pattern is $\sin\theta \cdot |\cos(kh\cos\theta)|$; maxima of the array factor occur when $kh\cos\theta = m\pi$.
Step 3. At $\theta = 60°$: $kh \cdot \frac{1}{2} = m\pi$. $m = 0$ gives no $60°$ maximum, so $m = 1$: $kh = 2\pi \Rightarrow h = \lambda$.

GATE 2020 | Q.44 | 2 marks

In the voltage regulator shown below, V_I is the unregulated input at 15 V. Assume $V_{BE} = 0.7$ V and the base current is negligible for both the BJTs. If the regulated output V_O is 9 V, the value of R_2 is _____ Ω.

output V_O is 9 V, the value of R_2 is _____ Ω.



Answer: 800 [answer matches official key]

Solution. Step 1. The Zener sets the emitter of the error transistor at 3.3 V, so its base is at $3.3 + 0.7 = 4.0$ V.

Step 2. The base is fed from the divider $R_1 = 1$ kΩ, R_2 across V_O: $V_O \dfrac{R_2}{R_1 + R_2} = 4$.

Step 3. $9R_2 = 4(1000 + R_2) \Rightarrow 5R_2 = 4000 \Rightarrow R_2 = 800\ \Omega$.

GATE 2020 | Q.45 | 2 marks

The magnetic field of a uniform plane wave in vacuum is given by $\vec{H}(x, y, z, t) = (\hat{a}_x + 2\hat{a}_y + b\hat{a}_z)\cos(\omega t + 3x - y - z)$. The value of b is _____.

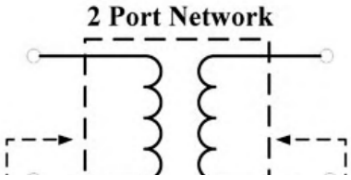
Answer: 1 [answer matches official key]

Solution. Step 1. For a plane wave $\vec{H} \perp \vec{k}$. From the phase, $\vec{k} \propto (3, -1, -1)$.

Step 2. $(1, 2, b) \cdot (3, -1, -1) = 3 - 2 - b = 0 \Rightarrow b = 1$.

GATE 2020 | Q.46 | 2 marks

For a 2-port network consisting of an ideal lossless transformer, the parameter S_{21} (rounded off to two decimal places) for a reference impedance of 10 Ω, is _____.



Answer: 0.8 [answer matches official key]

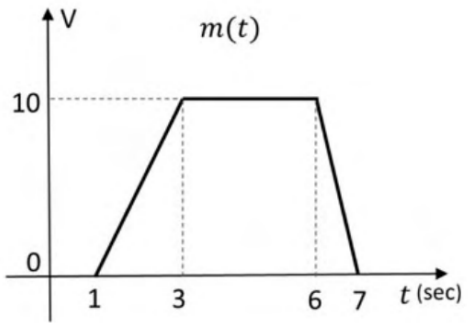
Solution. Step 1. For an ideal transformer of turns ratio $n : 1$ with equal reference impedances, $S_{11} = \dfrac{n^2 - 1}{n^2 + 1}$ and $S_{21} = \dfrac{2n}{n^2 + 1}$.

Step 2. With $n = 2$: $S_{21} = \dfrac{4}{5} = 0.8$.

GATE 2020 | Q.47 | 2 marks

$S_{PM}(t)$ and $S_{FM}(t)$ as defined below, are the phase modulated and the frequency modulated waveforms, respectively, corresponding to the message signal $m(t)$ shown in the figure. $S_{PM}(t) = \cos\left(1000\pi t + K_p m(t)\right)$ and $S_{FM}(t) = \cos\left(1000\pi t + K_f \int_{-\infty}^{t} m(\tau)d\tau\right)$, where K_p is the phase deviation constant in radians/volt and K_f is the frequency deviation constant in radians/second/volt. If the highest instantaneous frequencies of $S_{PM}(t)$ and $S_{FM}(t)$ are same, then the value of the ratio $\dfrac{K_p}{K_f}$ is _____ seconds.

deviation constant in radians/second/volt. If the highest instantaneous frequencies of $S_{PM}(t)$ and $S_{FM}(t)$ are same, then the value of the ratio $\frac{K_p}{K_f}$ is _____ seconds.



Answer: 2 [answer matches official key]

Solution. Step 1. Instantaneous angular frequency of PM: $1000\pi + K_p\, m'(t)$; of FM: $1000\pi + K_f\, m(t)$.
Step 2. From the figure $m(t)$ rises from 0 to 10 V between $t = 1$ and $t = 3$: maximum slope $m'_{max} = 5$ V/s; the maximum value is $m_{max} = 10$ V.
Step 3. Equal highest frequencies: $5K_p = 10K_f \Rightarrow \dfrac{K_p}{K_f} = 2$ s.

GATE 2020 | Q.48 | 2 marks

In a digital communication system, a symbol S randomly chosen from the set $\{s_1, s_2, s_3, s_4\}$ is transmitted. It is given that $s_1 = -3$, $s_2 = -1$, $s_3 = +1$ and $s_4 = +2$. The received symbol is $Y = S + W$. W is a zero-mean unit-variance Gaussian random variable and is independent of S. P_i is the conditional probability of symbol error for the maximum likelihood (ML) decoding when the transmitted symbol $S = s_i$. The index i for which the conditional symbol error probability P_i is the highest is _____.

Answer: 3 [answer matches official key]

Solution. Step 1. ML decision boundaries are the mid-points: -2, 0 and 1.5.
Step 2. Decision regions: $s_1 : (-\infty, -2)$, $s_2 : (-2, 0)$, $s_3 : (0, 1.5)$, $s_4 : (1.5, \infty)$.
Step 3. The outer symbols err on one side only; the inner symbols on both sides. s_2 has a margin of 1 on each side; s_3 has margins 1 and 0.5, so $P_3 = Q(1) + Q(0.5)$ is the largest. Hence $i = 3$.

GATE 2020 | Q.49 | 2 marks

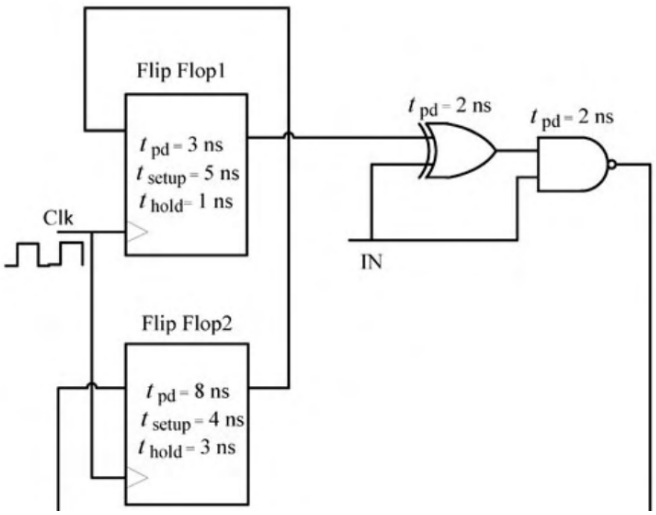
A system with transfer function $G(s) = \dfrac{1}{(s+1)(s+a)}$, $a > 0$ is subjected to an input $5\cos 3t$. The steady state output of the system is $\dfrac{1}{\sqrt{10}}\cos(3t - 1.892)$. The value of a is _____.

Answer: 4 [answer matches official key]

Solution. Step 1. Amplitude: $5|G(j3)| = \dfrac{5}{\sqrt{1+9}\sqrt{9+a^2}} = \dfrac{1}{\sqrt{10}} \Rightarrow \sqrt{9+a^2} = 5 \Rightarrow a = 4$.
Step 2. Check the phase: $-\tan^{-1} 3 - \tan^{-1} \dfrac{3}{4} = -71.57^\circ - 36.87^\circ = -108.4^\circ = -1.892$ rad ✓.

GATE 2020 | Q.50 | 2 marks

For the components in the sequential circuit shown below, t_{pd} is the propagation delay, t_{setup} is the setup time, and t_{hold} is the hold time. The maximum clock frequency (rounded off to the nearest integer), at which the given circuit can operate reliably, is _____ MHz.

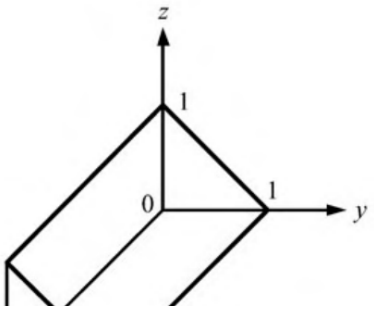


Answer: 77 [answer matches official key]

Solution. Step 1. The clock period must exceed the longest register-to-register path $t_{pd} + t_{logic} + t_{setup}$.
Step 2. Path $FF_2 \to FF_1$: $t_{pd2} + t_{setup1} = 8 + 5 = 13$ ns. The path through the XOR and NAND gates into FF_2 is $3 + 2 + 2 + 4 = 11$ ns, which is shorter.
Step 3. $T_{min} = 13$ ns, so $f_{max} = \dfrac{1}{13 \text{ ns}} = 76.9 \approx 77$ MHz.

GATE 2020 | Q.51 | 2 marks

For the solid S shown below, the value of $\iiint_S x\,dx\,dy\,dz$ (rounded off to two decimal places) is _____.



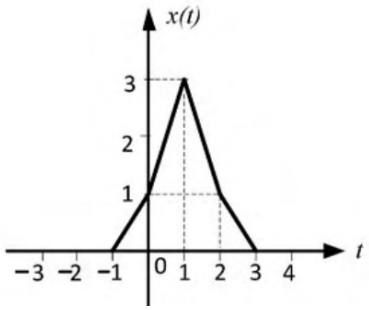
Answer: 2.25 [answer matches official key]

Solution. Step 1. The solid is a prism whose uniform cross-section in the y–z plane is the right triangle with legs 1 and 1 (area $\frac{1}{2}$), extending along x from 0 to 3.
Step 2. $\iiint x\,dV = \left(\int_0^3 x\,dx\right) \times (\text{cross-section area}) = \dfrac{9}{2} \times \dfrac{1}{2} = 2.25$.

GATE 2020 | Q.52 | 2 marks

$X(\omega)$ is the Fourier transform of $x(t)$ shown below. The value of $\displaystyle\int_{-\infty}^{\infty} |X(\omega)|^2 d\omega$ (rounded off to two decimal places) is _____.

$\int_{-\infty}^{\infty}|X(\omega)|^2\ d\omega$ **(rounded off to two decimal places) is** _____ .



Answer: 58.64 [answer matches official key]

Solution. Step 1. Parseval: $\int |X(\omega)|^2 d\omega = 2\pi \int x^2(t)\, dt$.

Step 2. $x(t)$ is piecewise linear through $(-1,0),(0,1),(1,3),(2,1),(3,0)$. Segment energies: $\int_{-1}^{0}(1+t)^2dt = \frac{1}{3}$; $\int_{0}^{1}(1+2t)^2dt = \frac{13}{3}$; $\int_{1}^{2}(5-2t)^2dt = \frac{13}{3}$; $\int_{2}^{3}(3-t)^2dt = \frac{1}{3}$.

Step 3. Total $= \dfrac{28}{3}$, so the integral is $2\pi \cdot \dfrac{28}{3} = 58.64$.

GATE 2020 | Q.53 | 2 marks

The transfer function of a stable discrete-time LTI system is $H(z) = \dfrac{K(z-\alpha)}{z+0.5}$, where K and α are real numbers. The value of α (rounded off to one decimal place) with $|\alpha| > 1$, for which the magnitude response of the system is constant over all frequencies, is _____.

Answer: -2.0 [answer matches official key]

Solution. Step 1. An all-pass section has its zero at the reciprocal of its pole: $\alpha = \dfrac{1}{p}$ with $p = -0.5$.

Step 2. $\alpha = -2$ (indeed $|\alpha| > 1$). Check: $|e^{j\omega} + 2| = 2|e^{j\omega} + 0.5|$ for all ω, so $|H| = 2|K|$.

GATE 2020 | Q.54 | 2 marks

X is a random variable with uniform probability density function in the interval $[-2, 10]$. For $Y = 2X - 6$, the conditional probability $P(Y \le 7 \mid X \ge 5)$ (rounded off to three decimal places) is _____.

Answer: 0.300 [answer matches official key]

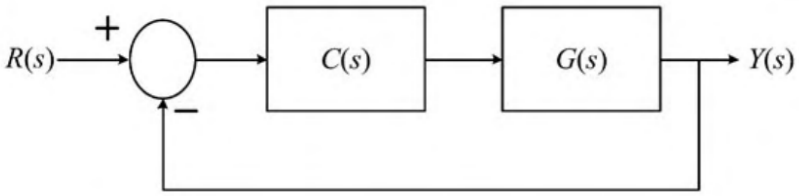
Solution. Step 1. $Y \le 7 \iff 2X - 6 \le 7 \iff X \le 6.5$.

Step 2. $P(5 \le X \le 6.5) = \dfrac{1.5}{12}$ and $P(X \ge 5) = \dfrac{5}{12}$.

Step 3. Conditional probability $= \dfrac{1.5}{5} = 0.3$.

GATE 2020 | Q.55 | 2 marks

Consider the following closed loop control system with $G(s) = \dfrac{1}{s(s+1)}$ and $C(s) = K\dfrac{s+1}{s+3}$. If the steady state error for a unit ramp input is 0.1, then the value of K is _____.



where $G(s)=\dfrac{1}{s(s+1)}$ and $C(s)=K\dfrac{s+1}{s+3}$. If the steady state error for a unit ramp input is 0.1, then the value of K is _____.

Answer: 30 [answer matches official key]

Solution. Step 1. Open loop: $C(s)G(s)=\dfrac{K(s+1)}{s(s+1)(s+3)}=\dfrac{K}{s(s+3)}$ (type 1).

Step 2. $K_v=\lim_{s\to 0} sC(s)G(s)=\dfrac{K}{3}$.

Step 3. $e_{ss}=\dfrac{1}{K_v}=\dfrac{3}{K}=0.1\Rightarrow K=30$.